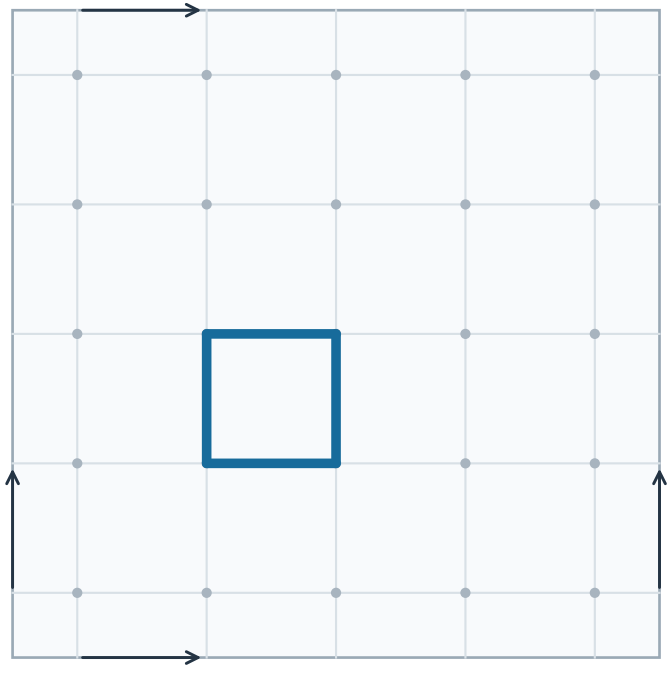
Closed loops on a torus

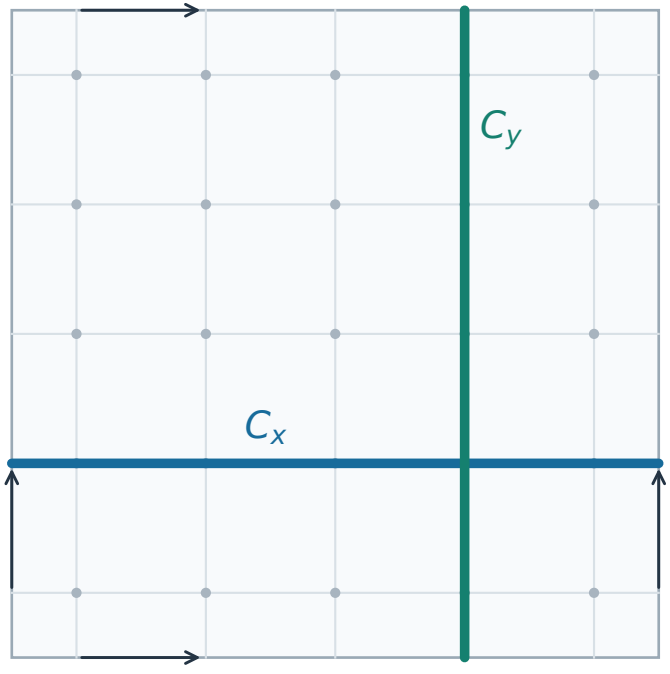
Identify opposite sides of the square, as indicated by matching arrows.

Contractible loop C_0



Can shrink by local plaquette flips.

Two noncontractible cycles



Wrap around the two directions of the torus.

A line reaching a seam continues at the matching point on the opposite side.

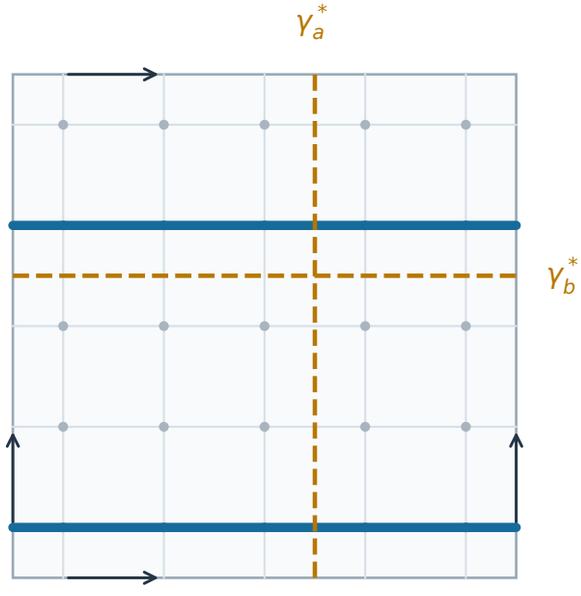
Every vertex has even occupancy. Four occupied edges at a vertex are allowed.

Four winding-parity sectors

$a = N(C \cap \gamma_a^*) \bmod 2, \qquad b = N(C \cap \gamma_b^*) \bmod 2$

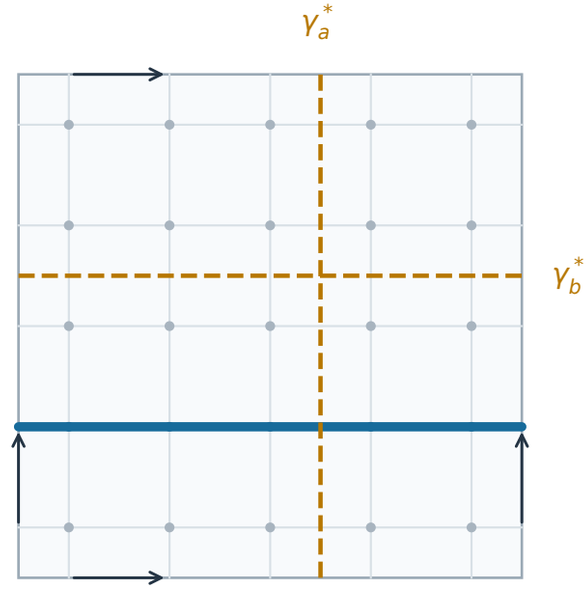
Occupied edges Dual measuring loops

$(a, b) = (0, 0)$



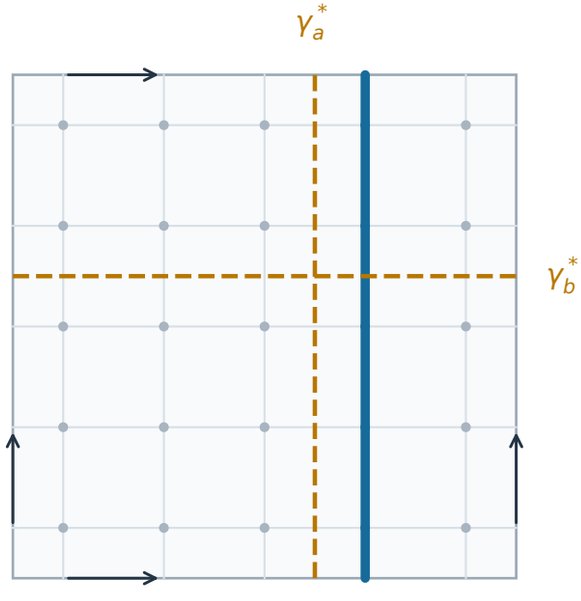
Crossings: (2, 0)

$(a, b) = (1, 0)$



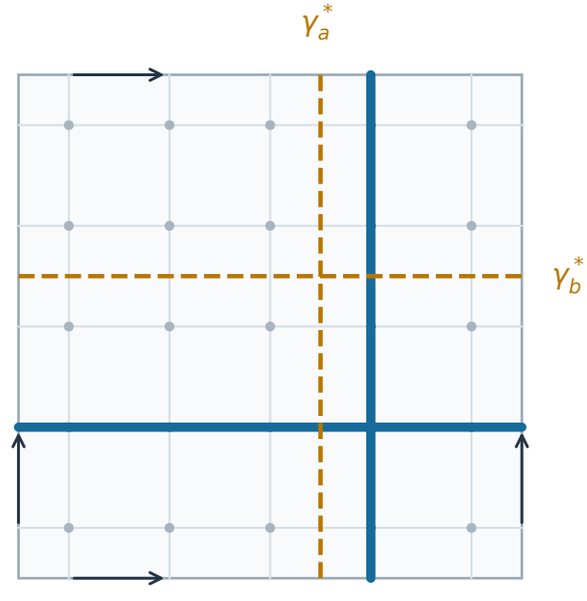
Crossings: (1, 0)

$(a, b) = (0, 1)$



Crossings: (0, 1)

$(a, b) = (1, 1)$



Crossings: (1, 1)

Each panel shows one configuration, not an entire ground state.

A sector ground state sums equally over all configurations with those parities.